



Solutions for metal industry drivelines

Benefits

- Optimized driveline performance and increased production
- Increased driveline and plant reliability
- Improved maintenance procedures

Typical applications

Industrial drivelines for:

- Conveyors
- Torpedo ladle cars
- Overhead cranes
- Converters
- Feeding tables
- Coiler/decoilers
- Rolling mill stands
- Straighteners/levelers
- Others

Increase reliability and profitability with SKF products and support

The motors, gearboxes and couplings that comprise drivelines used in rolling and steel mills provide power to heavy-duty processes, in environments that are often dirty, hot and dangerous.

In converters, very slow movements of small amplitude and long stationary periods – during which drivelines are subject to violent vibration – can seriously impact driveline performance. In rolling mills, drivelines may experience rapidly alternating working conditions such as sudden inversions between driving and braking actions, large torque fluctuations within short time periods, and constant torque in a wide speed range. Extremely high torque peaks can also occur, often exceeding driveline specifications.

All of these events impact heavily on driveline performance. Machine failure can result in reduced output, an increase in scrapped product, and – because driveline repair can be difficult and time-consuming – higher maintenance costs.

Rely on SKF for driveline reliability

SKF high performance products and support can help you to maximize driveline reliability, and contribute to increased productivity and reduced maintenance costs.

Bearings. SKF Explorer performance class bearings are well suited to the demands of gearboxes and motors in metals production. Available products in this execution range from taper roller bearings to deep groove ball bearings and self-aligning solutions. Large hybrid bearings are also available to prevent electric erosion in critical electrical motors.



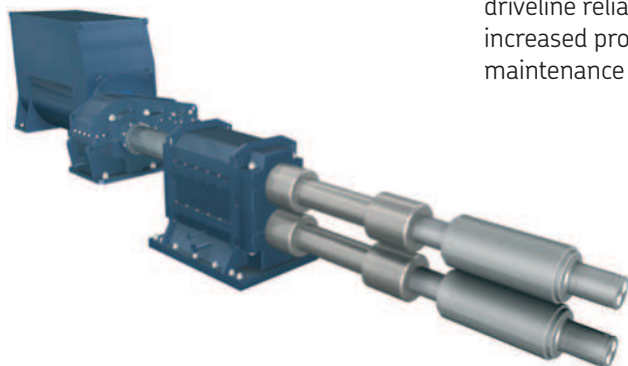
Sealing solutions. SKF radial shaft seals, combined with SKF SPEEDI-SLEEVE or wear sleeves for an optimal counterface, protect machinery in harsh operating environments. This total sealing system provides longer gear unit service life and allows for on-site replacement, reducing maintenance time and costs.

Maintenance and reliability services.

From basic condition monitoring to predictive maintenance and operator driven reliability, SKF can implement ongoing reliability programs, or equip and train on-site maintenance teams.

More solutions for optimized reliability.

SKF provides a variety of additional products and services that can enhance driveline performance and reliability. These include alignment and mounting tools, couplings, and oil and grease lubrication systems.





Increase the return on your maintenance investment with SKF

The whole idea behind the SKF 360° Solution is to help you get more out of your plant machinery and equipment investment. This may mean lowering your maintenance costs, raising your productivity, or both! Here's an example of the SKF 360° Solution at work in the metals industry.

Rolling mill doubles MTBF with SKF engineered solutions for gearboxes

A rolling mill experienced problems with a gearbox used to drive a steel strip-pickling line feeding stand. The process operated continuously, producing high volumes, in a dirty and hot environment. The gearbox and couplings failed frequently due to shock loads caused by the impacts between the rolling stock and the work rolls.

Knowing that SKF could provide a total package, including application engineering, high-performance products and installation services, the customer asked SKF to provide a solution to eliminate the recurring problem.

After an analysis of the gearbox failure, SKF determined that the problem was an unsuitable bearing arrangement on one specific shaft of the gearbox. The arrangement consisted of a pair of imperial (inch) taper roller bearings. The reliability of these bearings was very low, and their maintenance and replacement procedures complex and time consuming. In addition, the bearings were highly expensive, and delivery time could be as long as a year.

SKF's proposed solution was a self-contained unit built around SKF Explorer spherical roller thrust bearings. The unit included bearings mounted inside a flanged sleeve that fit on the gearbox's shaft and on the casing. The solution resulted in



cost savings through decreased downtime and related maintenance, easier mounting and lower spares inventory. Mean time between failure (MTBF) doubled from one to two years, and the company achieved a total ROI of 325% over two years as shown in the below table.

Return of investment (ROI)* in a two year period

Parts and labour savings	€2 800
Elimination of unplanned downtime	€10 000
Gross savings	€12 800
Investment in SKF solution	€3 000
Net savings	€9 800
ROI calculation	325%

* All numbers are rounded off and based on customer estimates. Your particular cost savings may vary.

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